

Context-Based Belief Revision

Paper #242

Abstract

In credibility-limited (CL) belief revision, an agent may reject new information when it is not considered credible relative to the agent's current beliefs. A central postulate of CL revision requires that all consequences of a credible formula are also credible. In this work, we introduce context-based (CB) belief revision, a generalization of CL revision that relaxes this requirement. In CB revision, a formula may be credible because it strengthens one of its non-credible consequences by adding sufficient supporting context. This enables scenarios where a credible formula can have non-credible consequences, something not allowed within the CL framework. We develop an axiomatic foundation for CB revision, identify specific subclasses, establish representation theorems, and investigate the connections between CB revision, its subclasses, and CL revision.

1 Introduction

Belief change operators (Alchourrón, Gärdenfors, and Makinson 1985; Gärdenfors 1988; Katsuno and Mendelzon 1991; Fermé and Hansson 2011) are designed to incorporate new information into an agent's belief base. A central assumption in belief revision is the primacy of update principle, which holds that newly acquired information should take precedence over the agent's current beliefs. This principle is typically formalized by the success postulate (Alchourrón, Gärdenfors, and Makinson 1985), corresponding to (R1) in Katsuno and Mendelzon (1991), which requires the new information to be included in the revised belief set.

However, in many realistic scenarios, agents may be reluctant to change, especially when the incoming information is perceived as unreliable or untrustworthy. To capture this, research on non-prioritized belief revision (see (Hansson 1997) for a survey) has explored weakening the success postulate so that the agent is not always forced to accept new input. While some approaches, such as belief promotion (Schwind, Konieczny, and Marquis 2018), introduce intermediate belief change operations that fall between revision and contraction, a more widely accepted relaxation of the success postulate is the principle of *relative success*, which requires that the agent's beliefs remain unchanged when the new input is not deemed credible. This idea gives rise to the *credibility-limited* (CL) revision framework. In the context of finite propositional logic, this frame-

work has been formalized by adapting the standard Katsuno and Mendelzon (KM) approach (Katsuno and Mendelzon 1991) to account for credibility-limited assumptions (Booth et al. 2012). The framework developed in this paper builds upon this credibility-limited revision setting.

Let us introduce a simple example of a credibility-limited revision scenario, which will serve as a running example throughout the paper:

Example 1. *Suppose that Alice, who was working from home yesterday, has good reasons to believe that her colleague Bob went to the institute. She also believes that Bob had lunch with his colleagues at the institute's cafeteria. Let us denote Alice's beliefs by φ . Later, Alice hears from another colleague that Bob actually did not have lunch with his colleagues, but instead had lunch with the prime minister. Let this new piece of information be denoted by α , where $\alpha \models \neg\varphi$, i.e., it contradicts her current beliefs. Since Alice finds this information implausible or not credible, she decides to reject it, resulting in no change to her belief state: $\varphi \circ \alpha \equiv \varphi$.*

This scenario illustrates a typical case of credibility-limited revision, where new information is dismissed due to its lack of credibility from the agent's perspective.

One of the key principles underlying credibility-limited (CL) revision operators is the *success monotonicity* postulate (P5) (Booth et al. 2012). This postulate states that if a formula is considered credible, then all its logical consequences should also be considered credible. However, this requirement may not hold in certain reasonable scenarios, as illustrated by the following continuation of our example:

Example 1 (continued). *Along with the input α (that Bob had lunch with the prime minister), Alice also receives more specific information β , stating that the minister of research visited the institute yesterday. In other words, Alice is told that Bob had lunch with the prime minister because the prime minister visited the institute. This additional context β makes the conjunction $\alpha \wedge \beta$ credible for Alice, even though the statement α on its own remains implausible to her.*

This illustrates a situation where the more informative input $\alpha \wedge \beta$ is accepted, while its logical consequence α is not credible when taken in isolation, thereby violating success monotonicity.

In fact, in our example, a degenerate case is when Alice is simply told that she is wrong, i.e., she receives the input $\alpha = \neg\varphi$, merely asserting that her current beliefs are incorrect without any context or explanation. In this case, the success monotonicity postulate (P5) in CL revision would require that any input contradicting an agent's current beliefs, no matter how unsubstantiated, must be treated as non-credible if its logical consequences are not credible. This blanket rejection of contradictory input can be seen as overly rigid or unreasonable in certain applications.

To address this, we propose a weakening of (P5), allowing a formula to be credible even if some of its logical consequences are not. Concretely, we drop (P5) and adjust the standard postulates accordingly (such as by weakening the trichotomy postulate) and introduce two new conditions that better capture the contextual nature of the revision scenarios we aim to model.

The remainder of the paper is structured as follows. The next section introduces the necessary background on Katsuno-Mendelzon (KM) revision and CL revision. Section 3 presents the axiomatic characterization of our new class of context-based (CB) revision operators. Section 4 provides a representation theorem in the traditional style, using total preorders over worlds defined via a structure we call a CB faithful assignment, which generalizes the standard CL assignment framework. Section 5 introduces two natural subclasses of CB revision operators and presents their respective representation results and interrelations.

All proofs are provided in the supplementary material.

2 Preliminaries

We consider a propositional language $\mathcal{L}$ of formulae built up from a finite set of propositional variables P and the usual connectives. A world is a mapping from P to $\{0, 1\}$. The symbol $\models$ denotes logical entailment between formulae, $\equiv$ logical equivalence, $[\varphi]$ the set of models of the formula φ , $\mathcal{L}_*$ the set of consistent formulae, and Ω the set of all worlds. A total preorder $(W, \preceq)$ over a set of worlds $W \subseteq \Omega$ is a reflexive and transitive relation on Ω . Given a total preorder $(W, \preceq)$ and a set $W' \subseteq \Omega$, the set $\min(W', \preceq)$ denotes the worlds from $W' \cap W$ that are minimal according to $\preceq$, i.e., $\min(W', \preceq) = \{\omega \in W \cap W' \mid \forall \omega' \in W \cap W', \omega \preceq \omega'\}$. A belief change operator $\circ$ is a mapping associating two formulae with a formula.

Let us recall the standard set of KM postulates for belief revision:

Definition 1 (KM revision operator (Katsuno and Mendelzon 1991)). *A belief change operator $\circ$ is called a KM revision operator if for all formulae $\varphi, \psi, \alpha, \beta \in \mathcal{L}$,*

- (R1) $\varphi \circ \alpha \models \alpha$
- (R2) *If $\varphi \wedge \alpha \not\models \perp$ then $\varphi \circ \alpha \equiv \varphi \wedge \alpha$*
- (R3) *If $\alpha \not\models \perp$ then $\varphi \circ \alpha \not\models \perp$*
- (R4) *If $\varphi \equiv \psi$ and $\alpha \equiv \beta$ then $\varphi \circ \alpha \equiv \psi \circ \beta$*
- (R5) $(\varphi \circ \alpha) \wedge \beta \models \varphi \circ (\alpha \wedge \beta)$
- (R6) *If $(\varphi \circ \alpha) \wedge \beta \not\models \perp$ then $\varphi \circ (\alpha \wedge \beta) \models (\varphi \circ \alpha) \wedge \beta$*

These operators can be characterized in terms of total preorders over worlds. This is done by associating each KM revision operator with a *faithful assignment*:

Definition 2 (Faithful Assignment). *A faithful assignment is a mapping associating each formula $\varphi \in \mathcal{L}$ with a total preorder $\preceq_\varphi$ such that the following conditions are satisfied for all $\varphi, \psi \in \mathcal{L}_*$:*

1. $[\varphi] = \min(\Omega, \preceq_\varphi)$
2. *If $\varphi \equiv \psi$, then $\preceq_\varphi = \preceq_\psi$*

Theorem 1 ((Katsuno and Mendelzon 1991)). *A belief change operator $\circ$ is a KM revision operator if and only if there exists a faithful assignment $\varphi \mapsto (C_\varphi, \preceq_\varphi)$ such that for all $\varphi, \alpha \in \mathcal{L}$:*

$$[\varphi \circ \alpha] = \min([\alpha], \preceq_\varphi)$$

Credibility-limited revision generalize KM revision operators by relaxing the success postulate (R1), and are characterized by the following set of postulates:

Definition 3 (Credibility-limited revision operator (Booth et al. 2012)). *A belief change operator $\circ$ is called a credibility-limited (CL) revision operator if for all formulae $\varphi \in \mathcal{L}_*$, $\psi, \alpha, \beta \in \mathcal{L}$,*

- (P1) $\varphi \circ \alpha \models \alpha$ or $\varphi \circ \alpha \equiv \varphi$
- (P2) *If $\varphi \wedge \alpha \not\models \perp$ then $\varphi \circ \alpha \equiv \varphi \wedge \alpha$*
- (P3) $\varphi \circ \alpha \not\models \perp$
- (P4) *If $\varphi \equiv \psi$ and $\alpha \equiv \beta$ then $\varphi \circ \alpha \equiv \psi \circ \beta$*
- (P5) *If $\varphi \circ \alpha \models \alpha$ and $\alpha \models \beta$, then $\varphi \circ \beta \models \beta$*
- (P6) $\varphi \circ (\alpha \vee \beta) \equiv \begin{cases} \varphi \circ \alpha, & \text{or} \\ \varphi \circ \beta, & \text{or} \\ (\varphi \circ \alpha) \vee (\varphi \circ \beta) \end{cases}$

In CL revision, the success postulate (R1) is replaced by the relative success postulate (P1), which states that if the new input is not credible ($\varphi \circ \alpha \not\models \alpha$), the belief state should remain unchanged. Postulates (P2) and (P4) correspond to (R2) and (R4), and (P3) is a stronger, simplified form of (R3). The trichotomy postulate (P6) captures minimal change and corresponds to (R5) and (R6) in the KM framework, given (R1–R4) (Booth et al. 2012). The postulate (P5), as discussed earlier, plays a central role in CL revision. It requires that if a formula is credible (hence accepted) then all its logical consequences must also be credible. This condition has no direct counterpart in the KM framework, where it is trivially satisfied due to the success postulate (R1).

A common assumption in CL revision is that the agent's belief state is consistent, that is, $\varphi \in \mathcal{L}_*$. This assumption is harmless and aligns with the core principles of CL revision: non-credible inputs are rejected (P1), and the output of revision is always consistent (P3). Consequently, in the remainder of the paper, we assume that any belief revision operator $\circ$ is a function from $\mathcal{L}_* \times \mathcal{L}$ to $\mathcal{L}$.

CL revision operators can be semantically characterized by CL faithful assignments, which generalize the standard faithful assignments from the KM framework to accommodate the notion of input credibility:

Definition 4 (CL Faithful Assignment). A CL faithful assignment is a mapping associating each formula $\varphi \in \mathcal{L}_*$ with a pair $(C_\varphi, \preceq_\varphi)$, where $[\varphi] \subseteq C_\varphi \subseteq \Omega$, $\preceq_\varphi$ is a total preorder on C_φ , and such that the following conditions are satisfied for all $\varphi, \psi \in \mathcal{L}_*$:

1. $[\varphi] = \min(C_\varphi, \preceq_\varphi)$
2. If $\varphi \equiv \psi$, then $(C_\varphi, \preceq_\varphi) = (C_\psi, \preceq_\psi)$

Theorem 2 ((Booth et al. 2012)). A belief change operator $\circ$ is a CL revision operator if and only if there exists a CL faithful assignment $\varphi \mapsto (C_\varphi, \preceq_\varphi)$ such that for all $\varphi \in \mathcal{L}_*$, $\alpha \in \mathcal{L}$:

$$[\varphi \circ \alpha] = \begin{cases} \min([\alpha], \preceq_\varphi) & \text{if } [\alpha] \cap C_\varphi \neq \emptyset \\ [\varphi] & \text{otherwise} \end{cases}$$

CL faithful assignments reduce to standard faithful assignments precisely when $C_\varphi = \Omega$ for every $\varphi \in \mathcal{L}_*$, i.e., when all worlds are considered credible. As shown in (Booth et al. 2012), this implies that any CL revision operator satisfying (R1) is in fact a KM revision operator.¹

Theorem 3 ((Booth et al. 2012)). If CL revision operator $\circ$ satisfies (R1), then $\circ$ is a KM revision operator.

3 Context-Based Revision

Let us first introduce some vocabulary, which will prove useful to explain the rationale of the postulates we will introduce in this paper:

Definition 5 (Context-Based Credibility). Let $\circ$ be a belief change operator and $\varphi \in \mathcal{L}_*$. A formula $\alpha \in \mathcal{L}$ is:

- φ -absolutely credible for $\circ$ if for each $\beta \in \mathcal{L}$, $\varphi \circ (\alpha \vee \beta) \models \alpha \vee \beta$
- φ -absolutely non-credible for $\circ$ if for each $\beta \in \mathcal{L}$, $\varphi \circ (\alpha \wedge \beta) \not\models \alpha \wedge \beta$
- φ -contextually credible for $\circ$ if $\varphi \circ \alpha \models \alpha$ and there exists $\beta \in \mathcal{L}$ such that $\varphi \circ (\alpha \vee \beta) \not\models \alpha \vee \beta$
- φ -contextually non-credible for $\circ$ if $\varphi \circ \alpha \not\models \alpha$ and there exists $\beta \in \mathcal{L}$ such that $\varphi \circ (\alpha \wedge \beta) \models \alpha \wedge \beta$

We will omit “for $\circ$ ” (respectively, the prefix “ φ -”) when $\circ$ (respectively, φ) is clear from context. The four cases in Def. 5 are mutually exclusive and collectively exhaustive: every formula α falls into exactly one of them. A formula is said to be *credible* if it is absolutely or contextually credible, and *non-credible* otherwise. Intuitively, credible formulae are those accepted by the agent, either absolutely (when all its consequences are also credible) or otherwise contextually.

Note that (P5) can now be equivalently restated: it requires that all credible formulae be absolutely credible. In other words, CL revision rules out contextual credibility. Similarly, under (R1), KM operators accept all inputs and thus treat every formula as absolutely credible.

¹Technically, (R1) and (P3) are incompatible when $\alpha \models \perp$: (R1) requires $\varphi \models \alpha$, contradicting (P3). This issue is avoided by excluding this degenerate case and assuming consistent inputs.

Thus, in CL revision, credibility is non-contextual: no formula can be contextually credible or contextually non-credible. Our goal is to develop a framework for context-based (CB) revision that explicitly allows for contextual credibility. To do so, we weaken the CL framework by revisiting its postulates. (P1) governs revision by non-credible inputs. Since the agent should not revise when an input is non-credible, regardless of whether that non-credibility is absolute or contextual, we keep (P1) unchanged. (P2) captures minimal change when the input is consistent with the agent’s beliefs. In CB revision, this implies such inputs are absolutely credible, and revision should reduce to expansion. This is a natural assumption since it aligns with the agent’s current beliefs, so we retain (P2). (P3) and (P4), consistency and syntax irrelevance, are orthogonal to our generalization and are kept as is. (P5) is the key principle we relax. As discussed, it rules out contextually credible inputs, so we drop it entirely. (P6), the trichotomy postulate, reflects a principle of minimal change that we want to preserve in spirit. However, in its current form, it is too strong. For example, a direct consequence of (P6) is that if two formulae are credible, then their disjunction must also be credible. Equivalently, (P6) implies that:

Proposition 1. If a belief change operator $\circ$ satisfies (P6), then for each $\varphi \in \mathcal{L}_*$ and each $\alpha \in \mathcal{L}$, if $\varphi \circ \alpha \models \alpha$, then there is a complete formula $\phi_\omega \models \alpha$ such that $\varphi \circ \phi_\omega \not\models \phi_\omega$.

However, we can imagine a plausible scenario where every complete formula is credible for an agent (due to its precision or sufficient contextual support), yet the agent may still refuse to revise her beliefs, φ , upon receiving an ambiguous input, such as being simply told she is “wrong” (i.e., $\alpha = \neg\varphi$).

Example 2 (continued). In our introductory example, Alice may find it credible that the prime minister visited Bob’s institute yesterday (β) and also credible that the prime minister visited the town office (γ). However, the disjunction $\beta \vee \gamma$ might be too vague for Alice to determine a plausible world in which it holds. As a result, $\beta \vee \gamma$ becomes contextually non-credible for Alice, as it doesn’t specify which scenario (β or γ) actually occurred.

To accommodate such situations, we propose weakening the trichotomy postulate as follows:

(RT) If $\varphi \circ (\alpha \vee \beta) \models \alpha \vee \beta$, then

$$\varphi \circ (\alpha \vee \beta) \equiv \begin{cases} \varphi \circ \alpha & \text{or} \\ \varphi \circ \beta & \text{or} \\ (\varphi \circ \alpha) \vee (\varphi \circ \beta) & \end{cases}$$

We can observe that in presence of (P1), (P2) and (P3), the postulate (RT) is equivalent to credibility-limited weakenings of (R5) and (R6):

(CR56) If $\varphi \circ \alpha \models \alpha$ and $(\varphi \circ \alpha) \wedge \beta \not\models \perp$, then $\varphi \circ (\alpha \wedge \beta) \models (\varphi \circ \alpha) \wedge \beta$

Proposition 2. If $\circ$ is a belief change operator satisfying (P1), (P2) and (P3), then $\circ$ satisfies (RT) if and only if it satisfies (CR56).

Notably, (CR56) can be seen as a conjunction of the *guarded subexpansion* postulate from (Hansson et al. 2001) ((R6') If $\varphi \circ \alpha \models \alpha$ and $(\varphi \circ \alpha) \wedge \beta \not\models \perp$, then $\varphi \circ (\alpha \wedge \beta) \models (\varphi \circ \alpha) \wedge \beta$) and a similar weakening of (R5). While (Booth et al. 2012) showed that (P6) is equivalent to (R5) and (R6') in the presence of the remaining CL postulates, here we need to weaken (R5) similarly, since removing (P5) could lead to situations where $\varphi \circ (\alpha \wedge \beta) \models \alpha \wedge \beta$ and $\varphi \circ \alpha \not\models \alpha$, causing $(\varphi \circ \alpha) \wedge \beta \equiv \varphi \wedge \beta$ not to entail $\varphi \circ (\alpha \wedge \beta)$.

At this stage, we have only removed (P5) and weakened the trichotomy postulate (P6) into (RT). However, the resulting set (P1-P4) and (RT) is still insufficient to avoid some unexpected behaviors. One such behavior concerns when a contextually credible formula becomes non-credible when combined with another formula. In this case, the non-credibility of the conjunction should be absolute: its credibility should not be restored by adding further context. We introduce this as the first core principle for CB revision, which we call Non-Credibility Persistence (NCP):

(NCP) If $\varphi \circ \alpha \models \alpha$, $\varphi \circ (\alpha \vee \beta) \not\models \alpha \vee \beta$ and $\varphi \circ (\alpha \wedge \gamma) \not\models \alpha$, then $\varphi \circ (\alpha \wedge \gamma \wedge \delta) \not\models \alpha$

Accordingly, interpreting this condition using the vocabulary from Def. 5, (NCP) means that if a formula α is contextually credible and a strengthening of it, $\alpha \wedge \gamma$, is non-credible, then $\alpha \wedge \gamma$ is absolutely non-credible. In other words, when α is contextually credible, the non-credibility of a strengthened formula like $\alpha \wedge \gamma$ cannot be attributed to a lack of context support. Instead, it means that $\alpha \wedge \gamma$ is simply not credible in any further context: there is no credible situation where it holds. Expanding $\alpha \wedge \gamma$ with additional information (such as $\alpha \wedge \gamma \wedge \delta$) would not restore its credibility.

Let us illustrate the rationale of (NCP) using our running example:

Example 1 (continued). Suppose again that the fact that Bob had lunch with the prime minister yesterday (α) is not credible for Alice. However, with the additional context that the prime minister visited Bob's institute yesterday (β), the formula $\alpha \wedge \beta$ becomes contextually credible. Now, suppose Alice receives further input in conjunction with $\alpha \wedge \beta$: Bob had dinner with the prime minister yesterday (γ). While Alice may find the situation where Bob had dinner with the prime minister, supported by his/her visit to Bob's institute ($\alpha \wedge \gamma$), plausible, she may not accept the idea that Bob had both lunch and dinner with the prime minister. Therefore, $\alpha \wedge \beta \wedge \gamma$ is not credible. The reason for this is that Alice cannot conceive of any world where this scenario holds, meaning that any further restrictions or additional information will also make $\alpha \wedge \beta \wedge \gamma$ absolutely non-credible.

Our second postulate is the principle of Contextual Cohesion (CC), which stipulates that if two formulas α and β are individually credible, and if their conjunction $\alpha \wedge \beta$ is also credible, then their disjunction $\alpha \vee \beta$ must be credible:

(CC) If $\varphi \circ \alpha \models \alpha$, $\varphi \circ \beta \models \beta$ and $\varphi \circ (\alpha \wedge \beta) \models \alpha \wedge \beta$, then $\varphi \circ (\alpha \vee \beta) \models \alpha \vee \beta$

If any of the formulas α , β , or $\alpha \wedge \beta$ were absolutely credible, the disjunction $\alpha \vee \beta$ would trivially be credible as well.

Therefore, this postulate is particularly relevant for *contextually* credible inputs. According to the vocabulary introduced in Def. 5, (CC) requires that if two formulas are contextually credible and their conjunction is contextually credible, then their disjunction must also be contextually credible.

The rationale behind this condition is to ensure that credibility is maintained when combining information within the *same* context. If two pieces of information, α and β , are individually contextually credible, they are each supported by sufficient context. The added premise that $\alpha \wedge \beta$ is contextually credible indicates that these two pieces of information are not supported by separate contexts but are jointly credible within a *single*, cohesive context. In this case, their disjunction, $\alpha \vee \beta$, should also be credible because it does not introduce any ambiguity: it remains tied to the same context. Thus, this axiom ensures that credibility is not lost when combining information that already aligns within a context.

Example 1 (continued). Let us assume again that both $\alpha \wedge \beta$ and $\beta \wedge \gamma$ are contextually credible for Alice: because the prime minister visited Bob's institute yesterday, Alice finds it credible that Bob had lunch or dinner with the prime minister. Now, suppose Alice also finds it credible that Bob had both lunch and dinner with the prime minister, i.e., $\alpha \wedge \beta \wedge \gamma$ is also credible. In this case, the disjunction $\alpha \vee \gamma$, along with the context that the prime minister visited Bob's institute (β), should also be credible for Alice. Since both $\alpha \wedge \beta$ and $\beta \wedge \gamma$ are individually credible, and their conjunction $\alpha \wedge \beta \wedge \gamma$ is credible as well, both statements fall within a single context. And since in this context, either α or γ can be accepted in conjunction with β , the disjunction $\beta \wedge (\alpha \vee \gamma)$ does not introduce any ambiguity.

With these two additional axioms in place, we are now ready to introduce our Context-Based (CB) revision operator:

Definition 6. A belief change operator $\circ$ is called a Context-Based (CB) revision operator if it satisfies (P1-P4), (RT), (CC) and (NCP).

It is important to note that this class of operators generalizes the credibility-limited framework. Specifically, it weakens (P5) and nothing else:

Proposition 3. A belief change operator is a CL revision operator if and only if it is a CB revision operator that satisfies (P5).

We can show that the generalization is strict, by presenting a concrete example of CB operators that are not CL operators in the general case. Given two formulae $\mu, \sigma \in \mathcal{L}$, let $\bullet_{\mu, \sigma}$ to be the belief change operator defined for all formulae $\varphi \in \mathcal{L}_*$, $\alpha \in \mathcal{L}$ as:

$$\varphi \bullet_{\mu, \sigma} \alpha = \begin{cases} \varphi \wedge \alpha & \text{if } \varphi \wedge \alpha \not\models \perp \\ \mu \wedge \alpha & \text{if } \varphi \wedge \alpha \models \perp, \mu \wedge \alpha \not\models \perp, \\ & \text{and } \alpha \models \sigma \\ \varphi & \text{in the remaining case} \end{cases}$$

Proposition 4. For each $\mu, \sigma \in \mathcal{L}$, $\bullet_{\mu, \sigma}$ is a CB revision operator.

This operator is defined by two formulas, μ and σ . Intuitively, μ characterizes the set of credible worlds (for worlds

that are not already models of the beliefs φ), while σ represents a (single) context. For the operator $\bullet_{\mu,\sigma}$, a formula α is considered credible if either (1) α is consistent with the agent's current beliefs ($\varphi \wedge \alpha \not\models \perp$), or (2) it has some credible models according to μ ($\mu \wedge \alpha \not\models \perp$) and is sufficiently supported by the context σ ($\alpha \models \sigma$).

When the formula σ characterizing the context is valid, meaning no specific context is required for contextual credibility, every credible formula becomes absolutely credible. In this case, $\bullet_{\mu,\sigma}$ behaves as a CL operator. On the other hand, $\bullet_{\mu,\sigma}$ is not a CL revision operator in the general case, particularly when σ is not valid and the credible set is consistent with the context. For instance, if we take $\mu = \sigma = p \wedge q$, it is easy to verify that $\neg p \bullet_{\mu,\sigma} p \equiv \neg p$ and $\neg p \bullet_{\mu,\sigma} (p \wedge q) \equiv p \wedge q$. This example shows that (P5) is not satisfied by $\bullet_{\mu,\sigma}$. In fact, we can show that:

Proposition 5. *For each $\mu, \sigma \in \mathcal{L}$, the operator $\bullet_{\mu,\sigma}$ is:*

1. *a CL revision operator iff (σ is valid or $\mu \wedge \sigma \models \perp$)*
2. *a KM revision operator iff both μ and σ are valid*

The class of operators $\bullet_{\mu,\sigma}$ thus gives us a non-degenerate case of CB revision that does not fall within the CL framework.

4 A Representation Result

We now aim to demonstrate how CB operators can be semantically represented, following the standard approach in belief change, using total preorders over worlds. This is achieved by introducing the concept of a *CB assignment*, which associates each formula $\varphi \in \mathcal{L}_*$ with what we call a *context structure space*.

The core component of a context structure space is a tuple, referred to as a *context structure*, denoted by $\mathcal{C} = (Cont, Cred, \preceq)$, where $\emptyset \neq Cred \subseteq Cont \subseteq \Omega$, and $\preceq$ is a total preorder on $Cred$. A context structure $\mathcal{C} = (Cont, Cred, \preceq)$ is said to be *absolute* if $Cont = \Omega$. In the following, when a context is annotated with a superscript integer i , it implicitly refers to the context $\mathcal{C}_\varphi^i = (Cont_\varphi^i, Cred_\varphi^i, \preceq_\varphi^i)$.

A *context structure space* is a tuple $\mathcal{S} = \{\mathcal{C}^0, \dots, \mathcal{C}^k\}$, with $k \geq 0$, where $\mathcal{C}^0$ is an absolute context and for each $i, j \in \{1, \dots, k\}$, if $i \neq j$ then $Cred_\varphi^i \cap Cont_\varphi^j = \emptyset$.

Definition 7 (CB faithful Assignment). A CB faithful assignment (CB assignment for short) Ψ is a mapping associating each formula $\varphi \in \mathcal{L}_*$ with a context structure space $\mathcal{S}_\varphi = \{\mathcal{C}^0, \dots, \mathcal{C}^k\}$ where $k \geq 0$ and such that the following conditions are satisfied for all $\varphi, \psi \in \mathcal{L}_*$:

1. $[\varphi] = \min(Cred_\varphi^0, \preceq_\varphi^0)$
2. If $\varphi \equiv \psi$, then $\mathcal{S}_\varphi = \mathcal{S}_\psi$

We aim to demonstrate that CB revision operators can be represented in terms of CB assignments. To begin, let us first provide some intuition regarding the objects involved in a CB assignment and the conditions that govern them. In a context structure space $\mathcal{C} = (Cont, Cred, \preceq)$, the absolute context structure $\mathcal{C}_\varphi^0$ is intended to represent a set of

credible worlds, $Cred_\varphi^0$, which make the satisfied formulae absolutely credible. Therefore, $Cred_\varphi^0$ plays a similar role to the set of credible worlds, C_φ , in a CL faithful assignment (Def. 4). The remaining context structures, $\mathcal{C}_\varphi^i$ for $i \in \{1, \dots, k\}$, are particularly relevant within our CB framework. In such a context structure $\mathcal{C}_\varphi^i = (Cont_\varphi^i, Cred_\varphi^i, \preceq_\varphi^i)$, $Cred_\varphi^i$ represents the *set of credible worlds* within the scope of that context structure, and $Cont_\varphi^i$ is a *contextual set of worlds*. Each context structure serves to characterize contextually credible formulae: a formula is contextually credible with respect to a given context structure $\mathcal{C}_\varphi^i$ if all of its models fall within $Cont_\varphi^i$, and at least one of its models is consistent with the set $Cred_\varphi^i$ of credible worlds.

To make this notion of credible formulae with respect to a given context structure precise, let us proceed:

Definition 8. *Given a context structure $\mathcal{C} = (Cont, Cred, \preceq)$, we say that a formula $\alpha \in \mathcal{L}$ is $\mathcal{C}$ -credible if $[\alpha] \cap Cred \neq \emptyset$ and $[\alpha] \subseteq Cont$.*

Let us observe that:

Lemma 1. *Let $\mathcal{S} = \{\mathcal{C}^0, \dots, \mathcal{C}^k\}$ be a context structure space and α be a formula. If α is $\mathcal{C}^i$ -credible for some $i \in \{1, \dots, k\}$, then for all $j \in \{1, \dots, k\}$, if $i \neq j$ then $[\alpha] \cap Cred_\varphi^j = \emptyset$ and $[\alpha] \not\subseteq Cont_\varphi^j$.*

Lemma 1 tells us that for each formula α , α is $\mathcal{C}^i$ -credible for at most one non-absolute context $\mathcal{C}^i \in \mathcal{S}$, i.e., with $i \in \{1, \dots, k\}$. When there exists $i \in \{1, \dots, k\}$ such that α is $\mathcal{C}^i$ -credible, in the following we simply denote such index i by i_α , and we say that α is $\mathcal{C}^{i_\alpha}$ -credible.

Let us now define a belief change operator from a CB assignment:

Definition 9. *Let Ψ be a CB assignment $\varphi \mapsto \mathcal{S}_\varphi$ with $\mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$, $k \geq 0$, and let $\circ$ be a belief change operator. We say that $\circ$ is represented by Ψ if for all $\varphi \in \mathcal{L}_*$, $\alpha \in \mathcal{L}$,*

$$[\varphi \circ \alpha] = \begin{cases} \min([\alpha], \preceq_\varphi^0) & \text{if } \alpha \text{ is } \mathcal{C}_\varphi^0\text{-credible} \\ \min([\alpha], \preceq_\varphi^{i_\alpha}) & \text{if } \alpha \text{ is not } \mathcal{C}_\varphi^0\text{-credible} \\ & \text{and } \alpha \text{ is } \mathcal{C}_\varphi^{i_\alpha}\text{-credible} \\ [\varphi] & \text{in the remaining case} \end{cases}$$

We can prove that:

Proposition 6. *If a belief change operator $\circ$ is represented by a CB assignment, then $\circ$ is a CB revision operator.*

The proof takes advantage of the following lemma:

Lemma 2. *If a belief change operator $\circ$ is represented by a CB assignment $\varphi \mapsto \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$, $k \geq 0$, then for each $\varphi \in \mathcal{L}_*$, $\alpha \in \mathcal{L}$, if α is φ -contextually credible then α is not $\mathcal{C}_\varphi^0$ -credible and α is $\mathcal{C}_\varphi^{i_\alpha}$ -credible.*

This lemma shows that contextually credible formulae are deemed credible due to their coherence with one of the non-absolute context structures within the context space. Moreover, combined with Lemma 1, it follows that there is exactly one such context structure that validates a given contextually credible formula.

We now aim to construct an assignment $\Psi_\circ$ from a given belief change operator $\circ$, such that the resulting assignment

qualifies as a CB assignment if $\circ$ is a CB operator. Moreover, in that case, the assignment represents $\circ$ in accordance with Definition 9.

The construction of this assignment proceeds as follows:

Definition 10 (Assignment corresponding to $\circ$). *Given a belief change operator $\circ$, the assignment corresponding to $\circ$, denoted by $\Psi_\circ$, is an assignment associating each formula $\varphi \in \mathcal{L}_*$ with a set $\mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$, $k \geq 0$, where for each $i \in \{0, \dots, k\}$, $\mathcal{C}_\varphi^i = (Cont_\varphi^i, Cred_\varphi^i, \preceq_\varphi^i)$, $Cred_\varphi^i, Cont_\varphi^i \subseteq \Omega$ and $\preceq_\varphi^i$ is a binary relation on $Cred_\varphi^i$, all of which are defined from $\circ$ as follows, for each $\varphi \in \mathcal{L}_*$:*

- $Cont_\varphi^0 = \Omega$
- $Cred_\varphi^0 = \{\omega \in \Omega \mid \forall \omega' \in \Omega \ \varphi \circ \phi_{\omega, \omega'} \models \phi_{\omega, \omega'}\}$
- $\{Cont_\varphi^i \mid 1 \leq i \leq k\} = \max(\mathcal{E}, \subseteq)$
- $\forall i \in \{1, \dots, k\}, Cred_\varphi^i = \{\omega \in Cont_\varphi^i \mid \varphi \circ \phi_\omega \models \phi_\omega\}$
- $\forall i \in \{0, \dots, k\} \ \forall \omega, \omega' \in Cred_\varphi^i \ \omega \preceq_\varphi^i \omega' \text{ iff } \omega \in [\varphi \circ \phi_{\omega, \omega'}]$

where $\max(\mathcal{E}, \subseteq) = \{W \in \mathcal{E} \mid \forall W' \in \mathcal{E} \ W \not\subseteq W'\}$, and $\mathcal{E} = \{W \subseteq (\Omega \setminus Cred_\varphi^0) \mid \exists \omega \in W : \forall \omega' \in W \ \varphi \circ \phi_{\omega, \omega'} \models \phi_{\omega, \omega'}\}$.

For the sake of illustration, take again the CB revision operator $\bullet_{\mu, \sigma}$ introduced in the previous section, that is defined by two formulae $\mu, \sigma \in \mathcal{L}$. It is not difficult to verify that the following assignment, denoted by $\Psi_{\bullet_{\mu, \sigma}}$, corresponds to $\bullet_{\mu, \sigma}$:

Definition 11 (Assignment $\Psi_{\bullet_{\mu, \sigma}}$). *For each $\mu, \sigma \in \mathcal{L}$, let $\Psi_{\bullet_{\mu, \sigma}} : \varphi \mapsto \mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ be assignment defined for each $\varphi \in \mathcal{L}_*$ as:*

- if $\neg\sigma \models \varphi$ or $\mu \wedge \sigma \models \varphi$, then $\mathcal{S}_\varphi = (\mathcal{C}_\varphi^0)$, with $Cred_\varphi^0 = [\varphi \vee (\mu \wedge \sigma)]$ and $\forall \omega, \omega' \in Cred_\varphi^0, \omega \preceq_\varphi^0 \omega' \text{ iff } \omega \in [\varphi] \text{ or } \omega' \in [\neg\varphi]$
- otherwise, $\mathcal{S}_\varphi = (\mathcal{C}_\varphi^0, \mathcal{C}_\varphi^1)$, with $Cred_\varphi^0 = [\varphi]$, $Cont_\varphi^1 = [\neg\varphi \wedge \sigma]$, $Cred_\varphi^1 = [\neg\varphi \wedge \mu \wedge \sigma]$, and $\forall i \in \{0, 1\}$, $\preceq_\varphi^i = Cred_\varphi^i \times Cred_\varphi^i$.

Then, accordingly:

Proposition 7. *For each μ, σ , the assignment $\Psi_{\bullet_{\mu, \sigma}}$ corresponds to $\bullet_{\mu, \sigma}$.*

We can now show that:

Proposition 8. *If $\circ$ is a CB operator, then its corresponding assignment $\Psi_\circ$ is a CB assignment.*

The proof goes through a set of notable observations:

Lemma 3. *Let $\circ$ be a CB revision operator. Then the assignment $\varphi \mapsto \mathcal{S}_\varphi$ that represents $\circ$ satisfies the following conditions, for each $\varphi \in \mathcal{L}_*$, all $i, j \in \{0, \dots, k\}$, all $\omega, \omega' \in \Omega$ and each $W \subseteq \Omega$:*

1. $[\varphi] \subseteq Cred_\varphi^0$
2. $\emptyset \neq Cred_\varphi^i \subseteq Cont_\varphi^i$
3. if $W \cap Cred_\varphi^i \neq \emptyset$ and $W \subseteq Cont_\varphi^i$, then $\varphi \circ \phi_W \models \phi_W$
4. if $j \geq 1$ and $i \neq j$, then $Cred_\varphi^i \cap Cont_\varphi^j = \emptyset$

Lemma 3.1 says that every model within the absolute context structure is considered credible, regardless of any specific context. Lemma 3.2 states that each context structure defines a contextual set of worlds that includes all its credible worlds, and ensures that the set of credible worlds is non-empty. Lemma 3.3 articulates a central feature of CB assignments: a formula is deemed credible if it entails the contextual set $Cont_\varphi^i$ and remains consistent with the corresponding set of credible worlds $Cred_\varphi^i$. Finally, Lemma 3.4 clarifies that, in the case of non-absolute context structures, each world can be credible with respect to at most one such structure, reinforcing the contextual partitioning of credibility.

The last step of our representation result consists in showing that the construction of a CB assignment $\Psi_\circ$ wrt. a CB operator is faithful to the representation of $\circ$ by $\Psi_\circ$ in Def. 10:

Proposition 9. *If a belief change operator $\circ$ is a CB revision operator, then it is represented by $\Psi_\circ$.*

The proof takes advantage of the following results:

Lemma 4. *Let $\circ$ be a CB revision operator and $\Psi_\circ$ be its corresponding CB assignment. Then for all $\varphi \in \mathcal{L}_*$, $\alpha \in \mathcal{L}$,*

1. $\forall i \in \{0, \dots, k\}$, if α is $\mathcal{C}_\varphi^i$ -credible, then $[\varphi \circ \alpha] \subseteq Cred_\varphi^i$
2. if $\varphi \circ \alpha \models \alpha$, then $\exists i \in \{0, \dots, k\}$ s.t. α is $\mathcal{C}_\varphi^i$ -credible

Lemma 4.1 states that revising by a credible formula, whether absolutely or contextually credible, results in an output whose models are all credible relative to a specific context structure. Lemma 4.2 says that any credible formula must be $\mathcal{C}_\varphi^i$ -credible with respect to at least one context structure (absolute or not) from the context structure space.

Prop. 7 and 9 together mean that using the assignment $\Psi_{\bullet_{\mu, \sigma}}$ one can recover the definition of $\bullet_{\mu, \sigma}$ from its representation through Def. 9.

Prop. 6, 8 and 9 together provide us with a (correct and complete) representation result, stated in the following theorem:

Theorem 4. *A belief change operator $\circ$ is a CB revision operator if and only if there exists a CB assignment Ψ associating each formula $\varphi \in \mathcal{L}_*$ with a context structure space $\mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ where $k \geq 0$, and such that for all $\varphi \in \mathcal{L}_*$, $\alpha \in \mathcal{L}$,*

$$[\varphi \circ \alpha] = \begin{cases} \min([\alpha], \preceq_\varphi^0) & \text{if } \alpha \text{ is } \mathcal{C}_\varphi^0\text{-credible} \\ \min([\alpha], \preceq_\varphi^i) & \text{if } \alpha \text{ is not } \mathcal{C}_\varphi^0\text{-credible} \\ & \text{and } \alpha \text{ is } \mathcal{C}_\varphi^i\text{-credible} \\ [\varphi] & \text{in the remaining case} \end{cases}$$

Before concluding this section, we illustrate how the CB assignment $\Psi_\circ$ representing a CB revision operator $\circ$ can be restricted to recover the characterizations of CL and KM revision operators. We also show how to capture CB operators that satisfy the original trichotomy condition (P6) satisfied by CL operators:

Proposition 10. *Let $\circ$ be a CB revision operator and $\Psi_\circ : \varphi \mapsto \mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ be its corresponding CB assignment. Then:*

1. $\circ$ is a CL revision operator iff $\forall \varphi \in \mathcal{L}_*, \mathcal{S}_\varphi = (\mathcal{C}_\varphi^0)$
2. $\circ$ is a KM revision operator iff $\forall \varphi \in \mathcal{L}_*, \text{Cred}_\varphi^0 = \Omega$
3. $\circ$ is satisfies (P6) iff $\forall \varphi \in \mathcal{L}_*, k \leq 1$

Prop. 10.1 and 10.2 provide alternative semantic characterizations of KM and CL revision in terms of CB assignments. Prop. 10.1 shows that a CB assignment represents a CL operator precisely when it contains only a single absolute context structure. In such cases, the CB assignment coincides with the structure $(\Psi_\varphi, \preceq_\varphi)$ used in a CL faithful assignment (see Def. 4). Similarly, Prop. 10.2 says that KM operators correspond to CB assignments with no non-absolute context structures and no non-credible worlds, mirroring the notion of faithful assignments as in Def. 2. Finally, Prop. 10.3 offers a noteworthy insight: CB operators satisfying the (non-credibility-limited) trichotomy condition (P6) of CL revision are exactly those whose CB assignment contains at most one context structure.

5 Two Classes of CB Revision Operators

This section introduces further constraints on CB revision, aimed at capturing more specific behaviors that may be relevant in certain applications. To this end, we define two subclasses of CB revision operators, each characterized by an additional postulate.

The first of these is the Context-Exclusive postulate ((CE)), which imposes a restriction on how CB revision handles contextually credible formulae. Specifically, it requires that such formulae exclude all worlds that are never deemed credible, i.e., any world that is not considered credible under any of the context structures in the assignment. This postulate enforces a stricter notion of credibility, ensuring that revision based on contextually credible information never reintroduces inherently non-credible worlds.

Definition 12 (Context-Exclusive revision operator). *A CB revision operator $\circ$ is called a Context-Exclusive (CE) operator if it satisfies the following condition:*

- (CE)** *If $\varphi \circ \alpha \models \alpha$, $\varphi \circ (\alpha \vee \beta) \not\models \alpha \vee \beta$ and $\alpha \wedge \gamma \not\models \perp$, then $\varphi \circ (\alpha \wedge \gamma) \models \alpha$*

We can paraphrase the Context-Exclusive postulate (CE) using the vocabulary introduced in Def. 5: it requires that if a formula α is (contextually) credible and the conjunction $\alpha \wedge \gamma$ is consistent, then this conjunction must also be credible. Intuitively, (CE) asserts that a formula is contextually credible only if all of its models correspond to situations that the agent considers credible. Consequently, if a formula is consistent but not credible, then restoring its credibility through weakening can only result in it becoming *absolutely* credible. In this sense, (CE) strengthens the (CC) postulate by enforcing a more rigorous standard of credibility preservation.

Example 2 (continued). *Suppose again that Alice initially finds it non-credible that Bob had lunch with the prime minister yesterday (α), but she considers it credible if this occurred because the prime minister visited Bob's institute ($\alpha \wedge \beta$). And assume that every possible situation entailing $\alpha \wedge \beta$ is considered credible by Alice. The additional*

context β renders the input $\alpha \wedge \beta$ “strongly” credible: every situation that satisfies it is deemed credible. It follows that if Alice later receives further evidence, such as Bob also having dinner with the prime minister ($\alpha \wedge \beta \wedge \gamma$), then, provided this conjunction is consistent, it too must be credible.

Finally, note that (P5) trivially implies (CE), since absolute credibility subsumes contextual credibility. As a result, CE operators still properly generalize the CL framework:

Proposition 11. *Every CL revision operator is a CE revision operator.*

The CE representation result can be found through the following proposition:

Proposition 12. *Let $\circ$ be a CB revision operator and $\Psi_\circ : \varphi \mapsto \mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ be its corresponding CB assignment. Then $\circ$ is a CE revision operator iff $\forall \varphi \in \mathcal{L}_*, \forall i \in \{1, \dots, k\}, \text{Cont}_\varphi^i = \text{Cred}_\varphi^i$.*

The following result shows that the generalization is strict, as some instances of the CB operators $\bullet_{\mu, \sigma}$ introduced earlier lead to CE operators that are not CL ones. It also shows that CE operators are a strict subclass of CB operators:

Proposition 13. *For each $\mu, \sigma \in \mathcal{L}$, the operator $\bullet_{\mu, \sigma}$ is a CE operator iff σ is valid or $\mu \wedge \sigma \models \perp$ or $\sigma \models \mu$.*

Indeed, Prop. 13 not only shows that CE operators is a strict subclass of the class of CB operators, but together with Prop. 5.1 it also shows that the class of CL operators is strictly included in the class of CE operators.

The second postulate we introduce is the Context-Sensitivity postulate (CS). This principle captures the intuition that the contextual credibility of a formula should not be undermined merely due to the existence of situations that are deemed non-credible in any context. In other words, if a formula is judged as not credible, it must be because it lacks adequate contextual support, rather than because it entails models that are globally non-credible. The (CS) postulate emphasizes that non-credibility should always be explainable in terms of insufficient context rather than due to the existence of unconditionally rejected worlds.

Definition 13 (Context-Sensitive revision operator). *A CB operator $\circ$ is called a Context-Sensitive (CS) operator if it satisfies the following condition:*

- (CS)** *If $\varphi \circ \alpha \models \alpha$ and $\forall \gamma \models \beta \varphi \circ \gamma \not\models \gamma$, then $\varphi \circ (\alpha \vee \beta) \models \alpha \vee \beta$*

Paraphrasing the Context-Sensitivity postulate (CS) using our vocabulary from Def. 5, it requires that contextually credible formulae α cannot lose their credibility when disjoined with absolutely non-credible formulae. In other words, the addition of non-credible situations to a contextually credible formula does not affect its credibility, as non-credible complete situations are irrelevant to the assessment of contextually credible formulae.

Example 1 (continued). *Suppose again that $\alpha \wedge \beta$ (Bob had lunch with the prime minister yesterday because he/she visited Bob's institute) is contextually credible for Alice. Now,*

consider the statement: “King Charles III visited Bob’s institute yesterday” (δ). Assume Alice finds δ absolutely non-credible, meaning no situation involving δ is acceptable to her. Now, consider the statement: “Bob had lunch with the prime minister yesterday, and either King Charles III or the prime minister visited Bob’s office” ($(\alpha \wedge (\beta \vee \delta))$). This statement is a weaker version of the contextually credible formula $\alpha \wedge \beta$ but adds an absolutely non-credible formula. Even with the addition of δ , the statement still retains enough contextual support for Alice to recognize the relevant context, so $(\alpha \wedge \beta) \vee \delta$ remains contextually credible for her.

Just as (P5) implies (CE) by vacuity, establishing that all CE operators are CL operators, we similarly observe that (P5) implies (CS) by vacuity, which shows that:

Proposition 14. *Every CL revision operator is a CS revision operator.*

An interesting observation is that the full CL framework is characterized by the class of CB operators satisfying (P6). The proof is based on Prop. 10.3 and the following lemma:

Lemma 5. *Let $\circ$ be a CS revision operator and $\Psi_\circ : \varphi \mapsto \mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ be its corresponding CB assignment. Then for each $\varphi \in \mathcal{L}_*$, $\{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ is such that $k \neq 1$.*

Proposition 15. *A belief change operator is CL operator iff it is a CS operator satisfying (P6).*

The CS representation result can be found through the following proposition:

Proposition 16. *Let $\circ$ be a CB operator and $\Psi_\circ : \varphi \mapsto \mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ be its corresponding CB assignment. Then $\circ$ is a CS revision operator iff $\forall \varphi \in \mathcal{L}_*$, $\forall i \in \{1, \dots, k\}$, $\text{Cont}_\varphi^i = \Omega \setminus \bigcup \{\text{Cred}_\varphi^j \mid j \in \{0, \dots, k\}, j \neq i\}$*

Prop. 16 states that for a CS operator $\circ$, all φ -credible situations ϕ_ω are included in every context set Cont_φ^i within the corresponding contextual structure space $\mathcal{S}_\varphi$ associated with $\circ$ through its CB assignment. This implies that a contextually non-credible formula α is non-credible because its set of models must overlap with at least two different context sets Cont_φ^i and Cont_φ^j , where $i \neq j$. Consequently, there exists a formula β such that $\alpha \wedge \beta$ is $\mathcal{C}_\varphi^{i_{\alpha \wedge \beta}}$ -credible ($[\alpha \wedge \beta] \subseteq \text{Cont}_\varphi^{i_{\alpha \wedge \beta}}$), and each model $\omega \in [\neg \alpha \wedge \beta]$ belongs to a different context set Cont_φ^i with $i \neq i_{\alpha \wedge \beta}$.

From Lemma 5 above, Prop.10.1, and Prop.14, we can deduce that every instance of a CB operator $\bullet_{\mu, \sigma}$ satisfying (CS) is necessarily a CL operator. More precisely:

Proposition 17. *For each $\mu, \sigma \in \mathcal{L}$, the operator $\bullet_{\mu, \sigma}$ is a CS operator iff it is a CL operator.*

This means that the class of CB operators $\bullet_{\mu, \sigma}$ cannot provide an example of a CS operator that is not also a CL operator. This is because an operator $\bullet_{\mu, \sigma}$ involves a single context defined by σ , and according to Lemma 5, at least two contexts are required to define non-degenerate CS operators.

Thus, we introduce a second class of concrete operators denoted by $\bullet_{\mu, \sigma}^{CS}$, which are also defined by two given for-

mulae $\mu, \sigma \in \mathcal{L}$ as follows, for each $\varphi \in \mathcal{L}_*$, $\alpha \in \mathcal{L}$:

$$\varphi \bullet_{\mu, \sigma}^{CS} \alpha = \begin{cases} \varphi \wedge \alpha & \text{if } \varphi \wedge \alpha \not\models \perp \\ \mu \wedge \alpha & \text{if } \varphi \wedge \alpha \models \perp, \mu \wedge \alpha \not\models \perp, \\ & \text{and } (\alpha \models \sigma \text{ or } \alpha \models \neg \sigma) \\ \varphi & \text{in the remaining case} \end{cases}$$

As intended, we get that:

Proposition 18. *For each $\mu, \sigma \in \mathcal{L}$, the operator $\bullet_{\mu, \sigma}^{CS}$ is a CS operator.*

For an operator $\bullet_{\mu, \sigma}^{CS}$, the formula μ defines the set of credible worlds (for worlds not already models of the beliefs φ), similar to an operator $\bullet_{\mu, \sigma}$. However, unlike $\bullet_{\mu, \sigma}$ operators, the formula σ here characterizes two distinct contexts: both the sets $[\sigma]$ and $[\neg \sigma]$ are used as context sets. This means that an input α is contextually credible only if either $\alpha \models \sigma$ or $\alpha \models \neg \sigma$.

By leveraging this new class of operators, we can now present a concrete example of a CS operator that is not a CL operator. Consider the case where $\mu = \sigma = p \wedge q$ (which is the same instantiation of μ and σ shown in Sec. 3 to provide an example of a CB operator that is not a CL operator). We can verify that $\neg p \bullet_{\mu, \sigma} p \equiv \neg p$ and $\neg p \bullet_{\mu, \sigma} (p \wedge q) \equiv p \wedge q$, demonstrating that (P5) is not satisfied by $\bullet_{\mu, \sigma}$. In fact, this instantiation can also be used to highlight an operator that does not satisfy (P5), especially when considering consistent formulas μ and consistent but non-valid formulas σ :

Proposition 19. *For each $\mu, \sigma \in \mathcal{L}$, the operator $\bullet_{\mu, \sigma}^{CS}$ is a CL operator iff $\sigma \models \perp$ or σ is valid or $\mu \models \perp$*

The last connection we investigate involves CB operators that are both CE and CS operators: they are simply called CES operators in the rest of this section. It is interesting to see that, as a direct consequence of the representations of CES operators in terms of their corresponding CB assignments (cf. Prop. 12 and 16), we get that:

Corollary 1. *Let $\circ$ be a CB revision operator and $\Psi_\circ : \varphi \mapsto \mathcal{S}_\varphi = \{\mathcal{C}_\varphi^0, \dots, \mathcal{C}_\varphi^k\}$ be its corresponding CB assignment. Then $\circ$ is a CES operator iff $\forall \varphi \in \mathcal{L}_*$, $\bigcup \{\text{Cred}_\varphi^i \mid i \in \{0, \dots, k\}\} = \Omega$*

Cor. 1 says that for a CES operator $\circ$, every world from Ω belongs to some (in fact, precisely one) credible set Cred_φ^i in the corresponding contextual structure space $\mathcal{S}_\varphi$ corresponding to $\circ$ through its CB assignment. This implies that every possible world is credible, or equivalently, that every non-credible statement can be made credible by adding some context support. Indeed:

Corollary 2. *A belief change operator $\circ$ is a CES operator iff it satisfies:*

(CES) *If α is a complete formula, then $\varphi \circ \alpha \models \alpha$*

Rephrasing the statement using the vocabulary from Def. 5, for CES operators, every formula is credible or contextually non-credible, or equivalently, there is no absolutely non-credible formula.

Another interesting consequence of our representation results is the interplay between CES operators and the trichotomy postulate (P6) satisfied by CL operators. Specif-

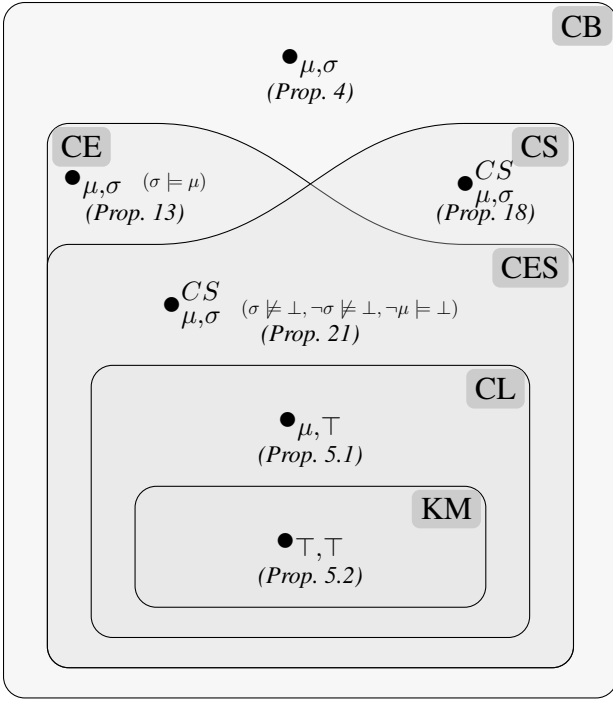


Figure 1: Dependencies between CB, CE, CS, CES, CL and KM operators. We have that $KM \subseteq CL$ (Th. 3), $CL \subseteq CS$ (Prop. 14), $CL \subseteq CE$, (Prop. 11), and $CE \subseteq CB$ and $CS \subseteq CB$ by definition. Each operator $\bullet_{\mu,\sigma}$, $\bullet_{\mu,\sigma}^{CS}$ in the figure represents a representative set of the class in which it is depicted, showing that each inclusion is strict.

ically, we can show that the full KM framework (for consistent inputs α) is entirely captured by CES operators that satisfy (P6).

Proposition 20. *A belief change operator is KM operator iff it is a CES operator that satisfies (P6).*

Lastly, we can show that CL operators form a strict subclass of CES operators. Specifically, we can instantiate an operator $\bullet_{\mu,\sigma}^{CS}$ such that it is a CES operator but not a CL operator. For example, the instantiation $\mu = \sigma = p \wedge q$ provided earlier in this section, which satisfies (CS) but not (P5), gives rise to an operator that also satisfies (CE). More precisely, we can show that:

Proposition 21. *For each $\mu, \sigma \in \mathcal{L}$, the CS operator $\bullet_{\mu,\sigma}^{CS}$ is a CE operator and not a CL operator iff $\sigma \not\models \perp$, $\neg\sigma \not\models \perp$, and μ is valid.*

We have now established the full connections between CB, CE, CS, CL, and KM operators. Additionally, we have demonstrated how adding (P6) imposes restrictions on each newly introduced class.

Fig. 1 summarizes some of these results, specifically highlighting the dependencies between these classes and illustrating how all classes (CB, CE, CS, CES, CL, and KM) are distinct. For each class C among CB, CE, CS, CES, CL, and KM, the figure depicts a concrete example of an operator (either $\bullet_{\mu,\sigma}$ or $\bullet_{\mu,\sigma}^{CS}$) that belongs to C and does not belong to any other class that does not includes C.

6 Summary and Discussion

The success condition in belief revision, captured by postulate (R1) in the KM axiomatization, has been a subject of controversy. A common weakening, relative success (P1), only accepts changes when the new input is among a set of credible formulas. This leads to the credibility-limited (CL) revision framework. In this paper, we introduced a strict generalization of CL revision, called context-based (CB) revision, by removing success monotonicity (P5) and weakening the trichotomy postulate (P6) to (RT). Success monotonicity requires that all consequences of a credible formula are also credible. In CB revision, a formula may be non-credible not because it has no credible world, but because it lacks sufficient context to support it. This means the formula may gain credibility by strengthening one of its non-credible consequences with context. We proposed two postulates, (CC) and (NCP), which model expected behaviors for CB revision as weakened versions of (P5). We provide a representation result for CB revision, in terms of total preorders over worlds and context sets, and introduce additional postulates, (CE) and (CS), to model specific CB situations. We also explore how these postulates interact, using CB assignment representations.

Although no work directly generalizes CL revision while considering contexts, a related concept is the role of explanations in belief revision, as explored in (Falappa, Kern-Isberner, and Simari 2002). In that work, explanations involve an explanans E (supporting beliefs) and an explanandum O (final conclusion). The framework operates on belief bases, requiring the explanans to be provided separately and entail the explanandum. In CB revision, we could interpret the added context as an explanans. However, since agents in our framework assume full rational inference (syntax independence, P4), we don't need to separate the explanans from the explanandum. In CB revision, adding context strengthens the formula ($O \wedge C$), whereas in (Falappa, Kern-Isberner, and Simari 2002), the explanation explicitly makes the relationship between E and O clear. In CB revision, context adds more information to support the formula's credibility, though it does not necessarily explain it.

Our work opens several research avenues. The two concrete operators we introduced, $\bullet_{\mu,\sigma}$ and $\bullet_{\mu,\sigma}^{CS}$, are simple and could be expanded to utilize the full expressivity of total pre-ordering. Additionally, distance-based revision operators, like Dalal's (Dalal 1988), could be used to define a broader class of CB operators. Another interesting direction is exploring how contexts evolve for iterative use, inspired by Darwiche and Pearl's epistemic states (Darwiche and Pearl 1997), as done in (Booth et al. 2012) in the CL framework. Further generalizations of CB revision are also possible, particularly in relaxing the vacuity postulate (P2) when contextual credibility is extended to belief expansion. For example, Alice's beliefs about Bob's lunch may not exclude the possibility of him dining with the prime minister, yet she may still not accept this due to insufficient context. Weakening (P2) to apply relative success in expansion would allow for greater flexibility, similar to our modification of (P6). This potential extension is left for future research.

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